COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010/Data Structures and Algorithms	**SECTION:**	CPE21S3
DATE PERFORMED:	07/07/2026	**DATE SUBMITTED:**	07/09/2026
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ACTIVITY NO. 1.1 **Algorithm, Flowchart, Pseudocode**

1. PROCEDURE OUTPUT

Code #1:

```
#include <iostream>
using namespace std;
int main(){
    int quiz1 = 0;
    int quiz2 = 0;
    int quiz3 = 0;
    int quizScr = 0;
    int quizTtl = 0;
    int quizAvr = 0;

    cout << "-----Quiz Score Display-----" << endl;
    cout << "Enter the overall number of scores: ";
    cin >> quizScr;

    cout << "Enter your Quiz #1 score: ";
    cin >> quiz1;

    cout << "Enter your Quiz #2 score: ";
    cin >> quiz2;

    cout << "Enter your Quiz #3 score: ";
    cin >> quiz3;

    quizTtl = quiz1 + quiz2 + quiz3;
    quizAvr = quizTtl / 2;
    cout << "Your overall quiz total: " << quizTtl << "/" << quizScr;
    cout << "\nYour quiz average is: " << quizAvr;

    return 0;
}
```

Output of the Code:

```
-----Quiz Score Display-----
Enter the overall number of scores: 75
Enter your Quiz #1 score: 20
Enter your Quiz #2 score: 24
Enter your Quiz #3 score: 23
Your overall quiz total: 67/75
Your quiz average is: 33
--------------------------------
Process exited after 7.082 seconds with return value 0
Press any key to continue . . .
```

Task #1 Code Analysis:

I first entered the values like the scores of the three quizzes, quiz score, average and the total using int. After that I used cout to display the score display and entered the quiz score of the three quizzes and using cin I gave it an interaction to get the score. To get the quiz total, I combined all the three quizzes and for average I divided the quiz total by 2 and display both the quiz total and quiz average using cout.

Code #2:

```cpp
#include <iostream>
using namespace std;
int main(){
    int price1 = 0;
    int price2 = 0;
    int price3 = 0;
    int pay = 0;
    int change = 0;
    int totalAmt = 0;

    cout << "------------CASHIERING SYSTEM-------------" << endl;
    cout << "Note: We only accept Philippine peso. " << endl;
    cout << "Other currencies are not available at the moment." << endl;
    cout << "               Thank you!        " << endl;

    cout << "\nEnter the price of the 1st item: PHP ";
    cin >> price1;

    cout << "Enter the price of the 2nd item: PHP ";
    cin >> price2;

    cout << "Enter the price of the 3rd item: PHP ";
    cin >> price3;

    totalAmt = price1 + price2 + price3;

    cout << "\nYour total amount: PHP " << totalAmt;

    cout << "\nPlease enter your desired payment: PHP ";
    cin >> pay;
    if (pay >= totalAmt){
        change = pay - totalAmt;
        cout << "Your change is: PHP " << change << endl;

        cout << "-------------TRANSCATIONS SCRIPT-----------" << endl;
        cout << "Item 1 Price: PHP " << price1;
        cout << "\nItem 2 Price: PHP " << price2;
        cout << "\nItem 3 Price: PHP " << price3;
        cout << "\n";
        cout << "\nTotal amount of all items: PHP " << totalAmt;
        cout << "\nTotal amount paid: PHP " << pay;
        cout << "\nTotal change: PHP " << change;
        cout << "\n";
        cout << "\nThank you for buying in our store. Support local! Have a nice day.";
    }else{
        cout << "\nInsufficient amount. Please enter the right desired amount.";
    }
    return 0;
}
```

Output of the Code:

If sufficient:	If not sufficient:

```
------------CASHIERING SYSTEM-------------
Note: We only accept Philippine peso.
Other currencies are not available at the moment.
               Thank you!

Enter the price of the 1st item: PHP 728
Enter the price of the 2nd item: PHP 526
Enter the price of the 3rd item: PHP 928

Your total amount: PHP 2182
Please enter your desired payment: PHP 2200
Your change is: PHP 18
-------------TRANSCATIONS SCRIPT-----------
Item 1 Price: PHP 728
Item 2 Price: PHP 526
Item 3 Price: PHP 928

Total amount of all items: PHP 2182
Total amount paid: PHP 2200
Total change: PHP 18

Thank you for buying in our store. Support local! Have a nice day.
--------------------------------
Process exited after 31.43 seconds with return value 0
Press any key to continue . . .
```

```
------------CASHIERING SYSTEM-------------
Note: We only accept Philippine peso.
Other currencies are not available at the moment.
               Thank you!

Enter the price of the 1st item: PHP 728
Enter the price of the 2nd item: PHP 526
Enter the price of the 3rd item: PHP 928

Your total amount: PHP 2182
Please enter your desired payment: PHP 2100

Insufficient amount. Please enter the right desired amount.
--------------------------------
Process exited after 12.61 seconds with return value 0
Press any key to continue . . .
```

Code Analysis on Task #2:

I first entered all the values like the three prices of the items, the change, total amount and the payment. I display the cashiering system, note and entering priced items using cout and let them interact using cin. After calculating the amount by combining the three prices, there would be an if/else statement if I have enough payment or not.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Pseudocode:

Task #1:

Task #2:

```
START
        INITIALIZE int price1 = 0
                   int price2 = 0
                   int price3 = 0
                   int pay = 0
                   int change = 0
                   Int totalAmt = 0

        DISPLAY "------------CASHIERING SYSTEM-------------"
        DISPLAY "Note: We only accept Philippine peso. "
        DISPLAY "Other currencies are not available at the moment."
        DISPLAY "              Thank you!        "

        DISPLAY "Enter the price of the 1st item: PHP "
        INPUT price1

        DISPLAY "Enter the price of the 2nd item: PHP "
        INPUT price2

        DISPLAY "Enter the price of the 3rd item: PHP "
        INPUT price3

        totalAmt = price1 +price2 + price3

        DISPLAY "Your total amount: PHP " totalAmt
```

```
START
        INITIALIZE int quiz1 = 0
                   int quiz2 = 0
                   int quiz3 = 0
                   int quizScr = 0
                   int quizTtl = 0
                   Int quizAvr = 0
        DISPLAY "-----Quiz Score Display-----"
        DISPLAY "Enter the overall number of scores: "
        INPUT quizScr

        DISPLAY "Enter your Quiz #1 score: "
        INPUT quiz1

        DISPLAY "Enter your Quiz #2 score: "
        INPUT quiz2

        DISPLAY "Enter your Quiz #3 score: "
        INPUT quiz3

        quizTtl = quiz1 + quiz2 + quiz3
        quizAvr = quizTtl / 2
        DISPLAY "Your overall quiz total: " quizTtl " / " quizScr
        DISPLAY "Your quiz average is: " quizAvr
STOP
```

```
        DISPLAY "Please enter your desired amount: PHP "
        INPUT pay
        IF/ELSE
        IF
                change = pay - totalAmt
                DISPLAY "Your change is: PHP " change
                DISPLAY "----------TRANSCATIONS SCRIPT----------"
                DISPLAY "Item 1 Price: PHP " price1
                DISPLAY "Item 2 Price: PHP " price2
                DISPLAY "Item 3 Price: PHP " price3
                DISPLAY "Total amount of all items: PHP " totalAmt
                DISPLAY "Total amount paid: PHP " pay
                DISPLAY "Total change: PHP " change
                DISPLAY "Thank you for buying in our store. Support locall Have a nice day."
        ELSE
                DISPLAY "Insufficient amount. Please enter the right desired amount."
STOP
```

Flowchart:

Task #1:



Task #2:
Submission

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **ChatGPT**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code Assistance**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used ChatGPT as a tool for code assistance on Task #2. It helped me understand if-else since I didn't remember anything about if-else and also the pseudocode.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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